

A Smartphone-Powered Mobility Intelligence System for Informal Public Transportation.

- Smartphone Sensor Fusion
- Zero Hardware / No Driver App
- Privacy by Design & Aggregation



1 THE PROBLEM

Knowing a transport route does not mean knowing whether a ride is actually available. Informal transit changes with time, demand, traffic, and weather.

? "Can I actually get a ride here right now?"

📍 "Where is the better place to wait?"

🕒 "How reliable is transport at this hour?"

📶 "Will I still have mobile signal during the trip?"

Existing static route maps cannot answer these dynamic real-world conditions.

2 BENEFICIARIES

Primary Users:
Students, daily commuters, night workers, delivery & gig-economy workers.

Secondary Beneficiaries:
Transport coops, schools, LGUs, researchers, & telco mobility planners.

3 THE SOLUTION

RouteSense uses sensors inside passengers' phones to learn how informal transit behaves over time through two intelligence layers:

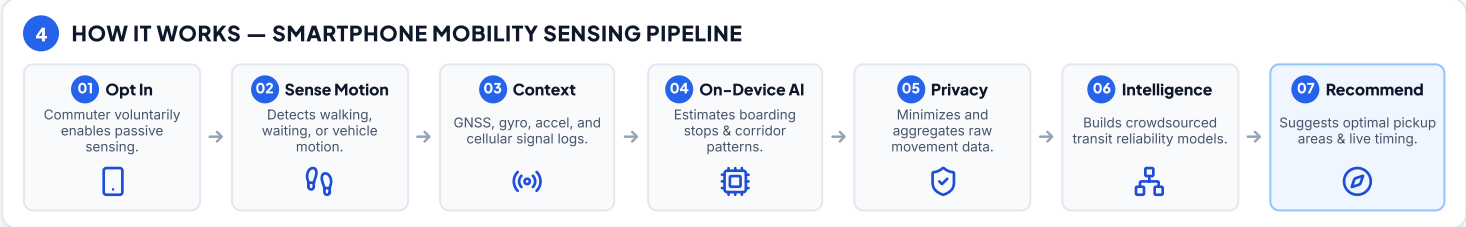
- StopSense:** Learns pickup areas, movement patterns, & service availability.
- SignalTrail:** Measures cellular signal quality along transit corridors.

STOPSENSE LIVE 8:30 PM

SERVICE PROBABILITY
88% HIGH

Typical Opportunity: 3-6 min

SignalTrail: ●●●●● Very Good



5 MOBILITY DECISION ENGINE

REAL-WORLD COMMUTER QUERY
"Going to Visayan Village — 8:30 PM"

RouteSense contrasts real-time availability, waiting range, confidence, and connectivity so commuters make informed choices.

Dynamic Opportunity Windows Signal-Aware Transit Choices

OPTION A
Main Pickup Area
FASTEST CHOICE

Service Probability	Typical Opportunity
88%	3-6 min
Confidence	SignalTrail
High (31 obs)	Moderate

OPTION B
Alternate Pickup Area
BEST CONNECTED

Service Probability	Typical Opportunity
74%	6-10 min
Confidence	SignalTrail
Moderate	Very Good

6 KEY FEATURES

Service Probability

Smart Pickup Areas

Opportunity Window

Time-Based Mobility

Confidence & Freshness

Signal-Aware Routes

Route Discovery

Offline Cached Data

No Driver App

7 EMERGING TECHNOLOGIES

Smartphone Sensor Fusion: Accel + Gyro + GNSS.
Activity Recognition: Walk → Wait → Vehicle.
Cellular Signal Intelligence: Network corridor maps.
On-Device Inference & Privacy Aggregation.
Geospatial Clustering: PostGIS + H3 Hexagonal Grid.

GPS/GNSS Accelerometer Gyroscope Cellular Modem Magnetometer

8 TECH STACK

React Native

SQLite

PostGIS + Uber H3

Kotlin Sensor Layer

MapLibre / OSM

Python / Pandas

TypeScript

Supabase / Postgres

XGBoost / SciKit

CCE 106



PROJECT PROPONENTS

LEAD

DEV

DEV

QA

Ryan Christian Deniega — Full-Stack & System Architect

Dominice Aseberos — Sensor Layer & Android Integration

Kyndel Roy Suarez — Geospatial & PostGIS Pipeline

Anjelo Vidal — Field Testing & ML Modeling

Course: **Application Development & Emerging Technologies (CCE)**

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[5] **Travel Behaviour & Society (2025)**. Passenger Smartphone Sensing for Transit Dynamics.